

# MADHUR BHAIYA

Jaipur, India

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## Summary

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Hands-on technology leader with 15+ years building production systems, now focused on **Gen AI applications**. Shipped a **natural-language-to-SQL** system on top of LLMs using **prompt engineering** (one-shot / few-shot exemplar selection), **retrieval-augmented generation (RAG)** over a schema catalog, and **semantic caching**. Strong backend (Golang, Python, PHP); deep **SQL/DBA** expertise on relational databases at scale (Postgres, MySQL, query optimization, replication, sharding, ProxySQL, distributed systems); full infrastructure ownership (Linux server administration, Nginx/Apache, Jenkins, GitHub Actions, Ansible). Built and led engineering teams from scratch across two ventures.

## Skills

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- **Gen AI / LLM**: Prompt engineering, one-shot and few-shot, retrieval-augmented generation (RAG), semantic caching, natural-language-to-SQL (NL2SQL), schema catalog modeling, LLM output guardrails / policy enforcement, self-correction loops, evaluation harnesses, Spec-Driven Development, LLM-assisted development workflows
- **Languages**: Go (Golang), Python, PHP, JavaScript, TypeScript, SQL, C++
- **Frameworks and Data**: ReactJS, Postgres, MySQL, ProxySQL, Qdrant (vector DB), Apache SOLR, Elasticsearch, RabbitMQ
- **Infrastructure and DevOps**: Linux, Nginx, Apache, Jenkins, GitHub Actions, Ansible, Git

## Experience

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### Chief Technology Officer

*KDK Softwares*

September 2020 – Present

*Jaipur, India*

- Designed and shipped a **natural-language-to-SQL** system for a multi-tenant tax-compliance SaaS — an LLM-powered interface that lets non-technical users ask queries on their tax data. Owned end-to-end: architecture, prompt design, retrieval pipeline, security/policy layer, evaluation, rollout.
- Built a **schema catalog** as the policy-and-context SSoT (table ownership, multi-tenant scopes, column metadata, value enumerations, join hints); per-request **RAG over Qdrant** (top-K retrieval + FK-chain anchor expansion) shrinks the per-request schema prompt from ~12k to ~3–5k tokens while preserving the static rules prefix for OpenAI prompt caching.
- Authored **sqlguard**, a custom Go AST-walker (vitesse sqlparser) that enforces 12+ policy rules on every LLM-emitted query before execution — SELECT-only, table whitelist, multi-scope **tenant-isolation verification** including FK-inherited tenancy via JOIN-tree walking, sensitive-column blocks, bind-parameter capture, LIMIT and MAX\_EXECUTION\_TIME injection. Defense-in-depth treating the LLM as untrusted; tenant identity is never read from the NL payload or LLM output.
- Implemented **few-shot exemplar selection** + a **semantic Q-to-SQL cache** in Qdrant (per-tenant filtered, sqlguard re-validates on every hit, promoted from thumbs-up feedback) plus a single-retry **self-correction loop** on eligible policy rejections. Built a structural **evaluation harness** that tracks every prompt/model revision as a measurable delta against a frozen golden set. Stack: Go, Qdrant, OpenAI.
- Rolled out **LLM-assisted development workflows** across a 25-engineer org centered on **Spec-Driven Development** — engineers author detailed, reviewable specs that LLMs use to generate consistent code and tests — shifting AI use from ad-hoc prompting to a repeatable, production-grade engineering practice.
- Founded the cloud SaaS engineering practice in a previously desktop-only company; cloud offerings now serve **10,000+ paying customers** across India. Built **five large-scale tax-compliance SaaS products** from scratch (ExpressGST, ExpressTDS, ExpressITR, ExpressReco, SpectrumCloud) plus internal CRM/admin systems. Led the **PHP (Laravel) to Golang** backend migration at production scale and upskilled engineers through the transition; hands-on across the stack — production Go and Python, deep SQL/DBA work (query optimization, replication, ProxySQL), React frontends, full DevOps ownership.

## Co-Founder and Director

Wholesalebox.in

April 2015 – September 2020

Jaipur, India

- Built the complete technology stack of **wholesalebox.in**, a B2B e-commerce platform (raised pre-Series A funding) — customer-facing web and mobile (separate codebases), Android and iOS apps, internal CRM and admin, seller-facing apps, B2C retailer storefronts.
- Scaled the business from zero to a **200+ employee team** including **40+ engineers**, with thousands of active retailers (customers) across India.
- Stack: PHP, MySQL, Postgres, ReactJS, React Native, Apache SOLR, Elasticsearch, RabbitMQ, Scala, Python, Nginx, Apache.

## Application Specialist, Remote Automation Solutions

Emerson (formerly Energy Solutions International Inc.)

April 2014 – July 2015

Calgary, Canada

- Programmed in a large-scale **C++ and Fortran** simulation codebase for real-time transient hydraulic modeling of oil and gas pipelines; object-oriented design with metaclasses backed by ObjectStore OODB.
- Built tooling in **Python** (network sockets, file management, fail-over wrappers) and **Java Swing / AWT** (GUI for hydraulic visualization); tuned production leak-detection models to reduce false-alarm rates.

## Education

### University of Alberta

M.Sc., Mechanical Engineering

January 2012 – December 2013

Edmonton, Canada | GPA: 3.9 / 4.0

- Dissertation: *An open-source two-phase non-isothermal mathematical model of a polymer electrolyte membrane fuel cell.*
- Mary Louise Imrie Graduate Student Award.

### Indian Institute of Technology Delhi (IIT Delhi)

B.Tech., Mechanical Engineering

August 2005 – May 2009

India | GPA: 7.2 / 10

## Earlier Experience and Research

- **Research Assistant, Energy Systems Design Lab, University of Alberta** (Jan 2012 – Mar 2014) — C++ finite-element model (30,000+ lines) for PEM fuel cell simulation; contributor to openfcst.org; Linux, MPI, CMake, TDD. Teaching Assistant for Thermo-Fluids and Energy Conversion.
- **Design Engineer, Hindustan Aeronautics Limited** (Jun 2009 – Dec 2011) — Aircraft fuel-system design; CFD and FEA; CAD in Unigraphics / Teamcenter for a turbo-trainer aircraft program. AFTC training Gold Medallist.
- **Intern, Whirlpool India** (2008) — Skin-condenser redesign with MATLAB thermal modeling; 31% efficiency improvement. **Undergraduate research** at IIT Delhi (coupled CFD/FEM, cascade refrigeration simulation) and FH Düsseldorf, Germany (harvester CFD).

## Certifications and Leadership

Pipeline Industry Professional Education (PIPE), 2014; AFTC Jalahalli — Gold Medallist; V.P. Communications, Mech. Eng. GSA, U. of Alberta; House Rep., Lit. & Dramatics Club, IIT Delhi.

## Publications

1. **Madhur Bhaiya**, Andreas Putz, Marc Secanell, “Analysis of non-isothermal effects on polymer electrolyte fuel cell electrode assemblies”, *Electrochimica Acta*, 167:160-171, 2014.
2. **Madhur Bhaiya**, Andreas Putz, Marc Secanell, “A comprehensive single-phase non-isothermal MEA model and analysis of non-isothermal effects”, *ECS Transactions*, 64(3):567-579, 2014.
3. Marc Secanell, Andreas Putz, Phil Wardlaw, Valentin Zingan, **Madhur Bhaiya**, Michael Moore, Jie Zhou, Chad Balen and Kailyn Domican, “OpenFCST: An open-source mathematical modelling software for polymer electrolyte fuel cells”, *ECS Transactions*, 64(3):655-680, 2014.

*Additional papers:* lead/co-author across *Hydrogen + Fuel Cells 2013* (Vancouver, Canada), *National Conference on Refrigeration and Air Conditioning (NCRAC)* 2009 (IIT Madras, Chennai), *ACRECONF* 2009 (New Delhi), and a FH Düsseldorf 2007 technical report on combine-harvester CFD.